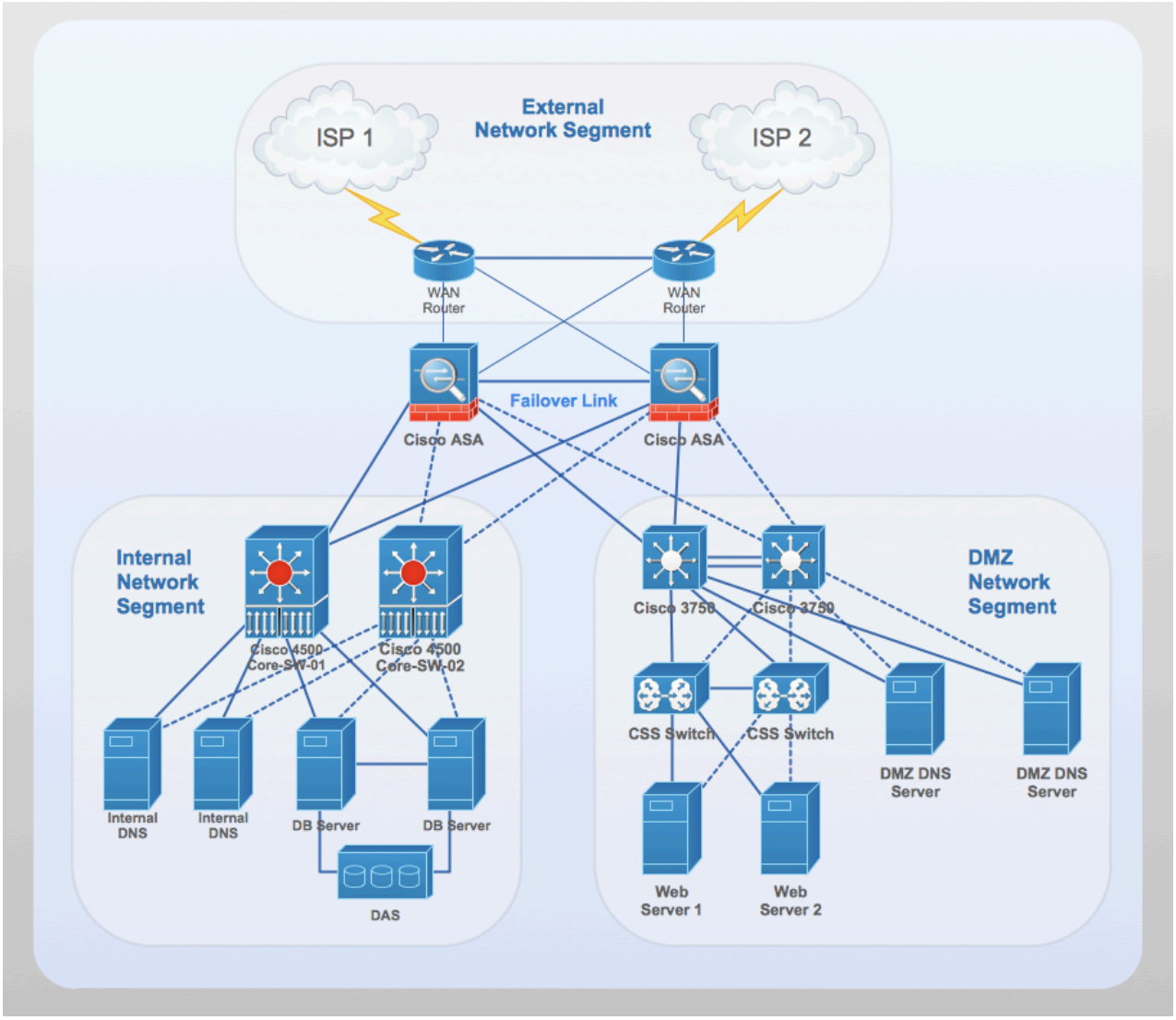
1. How the Internet Operates

Content
The Internet is a **global network of networks** that uses packet switching to transmit data between devices. When you request a webpage:

- Your device breaks the request into **small packets**, each labeled with source and destination IP addresses.
- **Routers** forward packets along optimal paths based on routing tables.
- Packets traverse multiple **Internet Service Providers (ISPs)** and backbone links (including submarine cables).
- The destination server reassembles packets and returns the response over the same packet-switched infrastructure.

Realtime Example
During the **October 2024 Meta outage**, millions of Facebook and Instagram users worldwide experienced service disruption when internal routing tables were misconfigured, causing packets to be dropped before reaching data centers.[1]

Reference Images
This enterprise network diagram illustrates ISP connections, failover links, core/edge routers, firewalls, and DMZ segments:



Network diagram showing Internet operation architecture including ISPs, routers, firewalls, internal and DMZ network segments with servers and switches.

Reference Videos

- "How the Internet Works in 5 Minutes" – YouTube: https://www.youtube.com/watch?v=7_LPdttKXPc
- "How does the Internet work? (Full Course)" – YouTube: https://www.youtube.com/watch?v=zN8YNNHcaZc

2. Understanding Internet Protocol & its Types

Content

An **Internet Protocol (IP)** defines how data packets are addressed and routed. Key protocol layers in the **TCP/IP suite** include:

- **IP (Internet Layer):** Routes packets using IP addresses.
- **TCP (Transport Layer):** Connection-oriented, reliable delivery.
- **UDP (Transport Layer):** Connectionless, low-latency delivery.

- **Application Layer Protocols:** HTTP/HTTPS (web), FTP/SFTP (file transfer), SMTP/IMAP/POP3 (email), DNS (name resolution).

Realtime Example

When you stream live video on YouTube, the platform uses **QUIC (HTTP/3)**—a UDP-based protocol—to reduce latency and recover faster from packet loss compared to TCP.

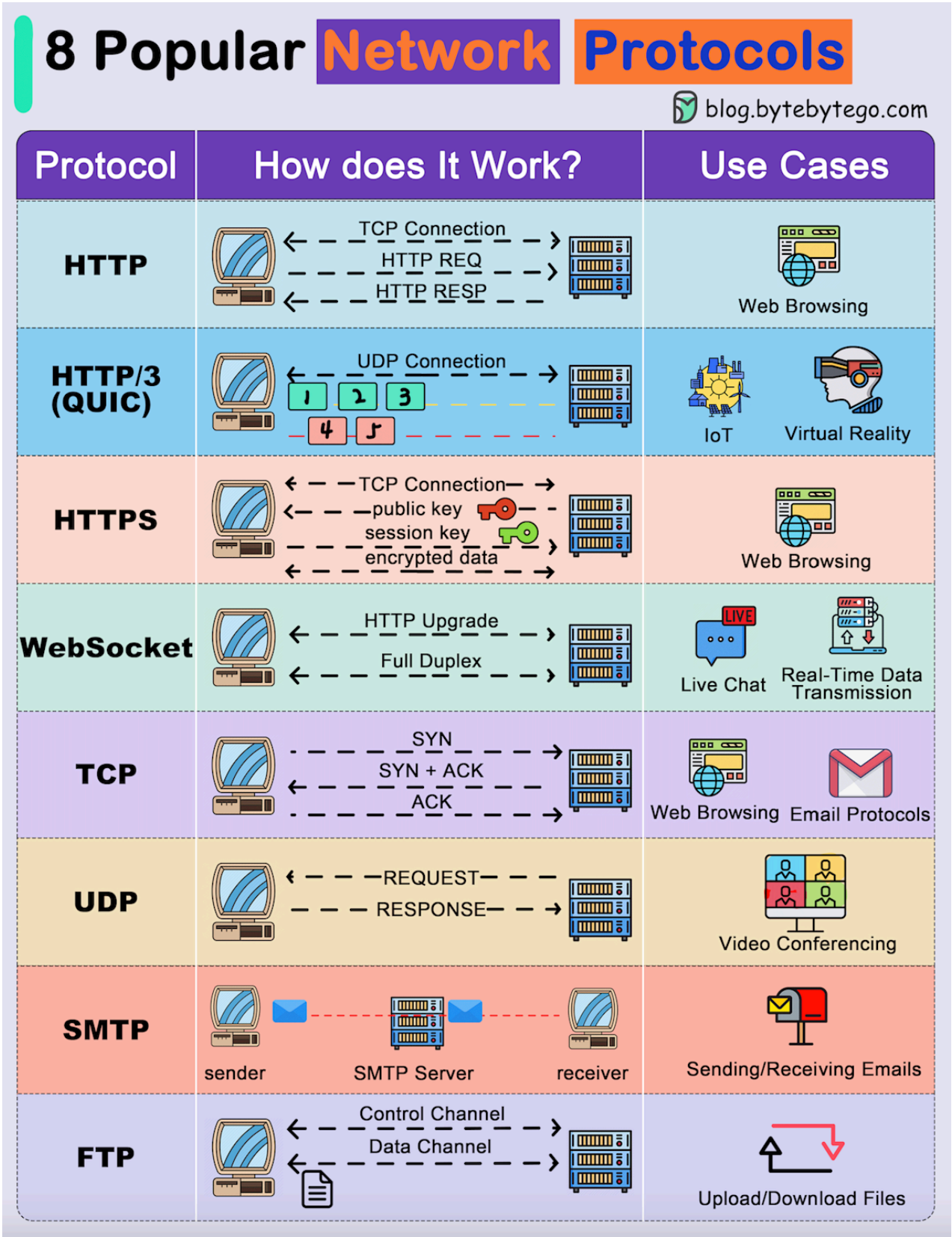


Diagram showing 8 popular network protocols with their operation methods and common real-world use cases.

Reference Images
Protocol stack diagram showing Application, Transport, Internet, and Link layers:

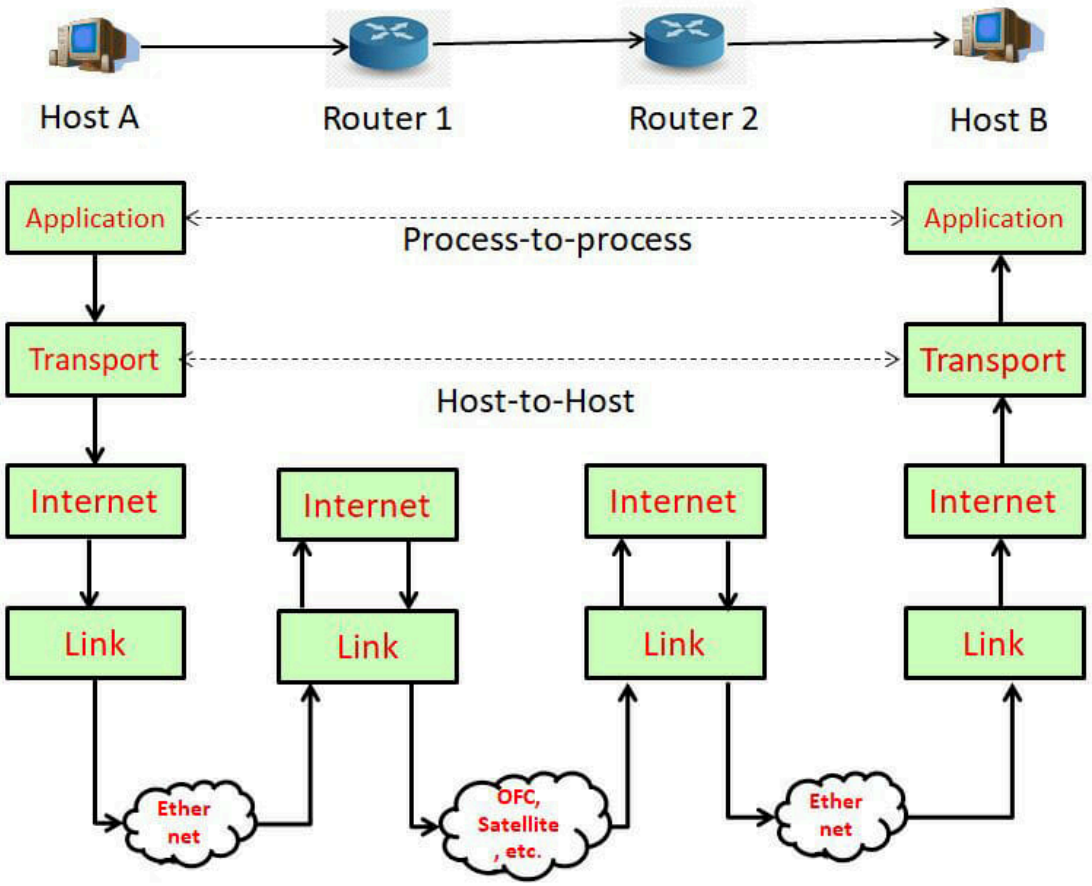


Diagram of the TCP/IP Internet protocol suite layers showing data flow between hosts via routers and different network media.

Reference Videos

- "Network Protocols Explained: Networking Basics" – YouTube: https://www.youtube.com/watch?v=1zVZ9cWFnCc
- "Types Of Network Protocol | TCP | IP | UDP | POP" – YouTube: https://www.youtube.com/watch?v=3NDhETVfrp0

3. OSI Model vs TCP/IP Model

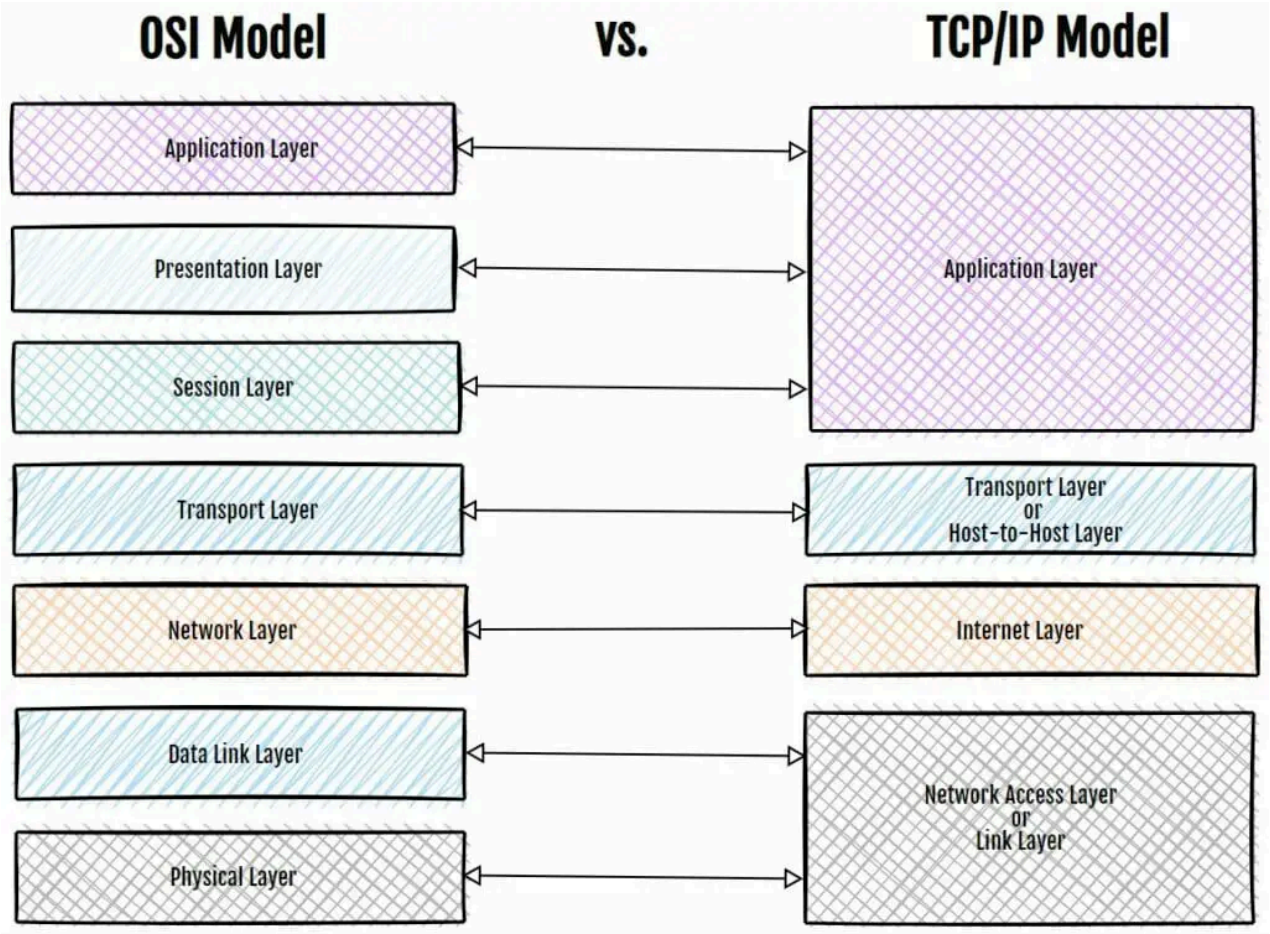
Content

- **OSI Model (7 layers):** Theoretical framework—Physical, Data Link, Network, Transport, Session, Presentation, Application.

- **TCP/IP Model (4 layers):** Practical suite—Network Access, Internet, Transport, Application.
- **Key Difference:** OSI separates presentation and session functions, while TCP/IP consolidates them in its Application layer.

Why It Matters
Network engineers use the OSI model for troubleshooting specific layers, while network implementations follow the TCP/IP suite on real hardware.

Reference Images
Comparison diagram of OSI vs TCP/IP layers:



Comparison of OSI model's seven layers with TCP/IP model's four layers showing equivalent functions between them.

Reference Videos

- "OSI & TCP/IP Models Explained in 9 minutes" – YouTube: https://www.youtube.com/watch?v=s4llaXGskm0
- "TCP/IP vs OSI: What's the Difference?" – YouTube: https://www.youtube.com/watch?v=tpgoQwMg__M

4. Internet Protocol Standards

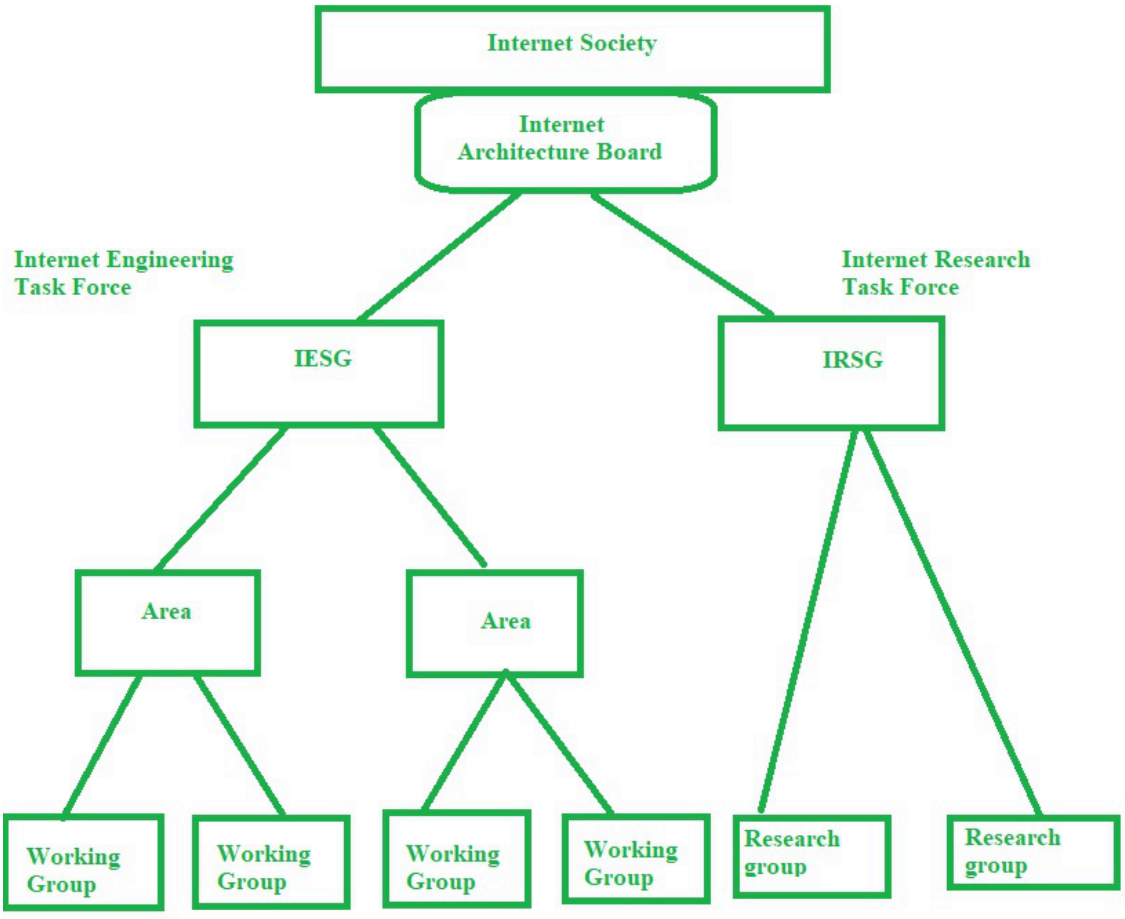
Content

Standards ensure **interoperability** across devices and vendors. The **IETF** publishes Internet standards via **RFCs**.

- **Internet Society (ISOC):** Oversees IETF governance.
- **Internet Architecture Board (IAB):** Provides technical oversight.
- **Internet Engineering Steering Group (IESG):** Approves RFCs for publication.

Realtime Example

When **HTTP/2** was standardized (RFC 7540), web servers and browsers adopted the multiplexing and header-compression enhancements to improve page load times.



Structure of Internet governance bodies responsible for internet protocol standards, including Internet Society, IETF, IRTF, and their working and research groups.

Reference Images

Organizational chart of IETF, IAB, IESG, and Working Groups:

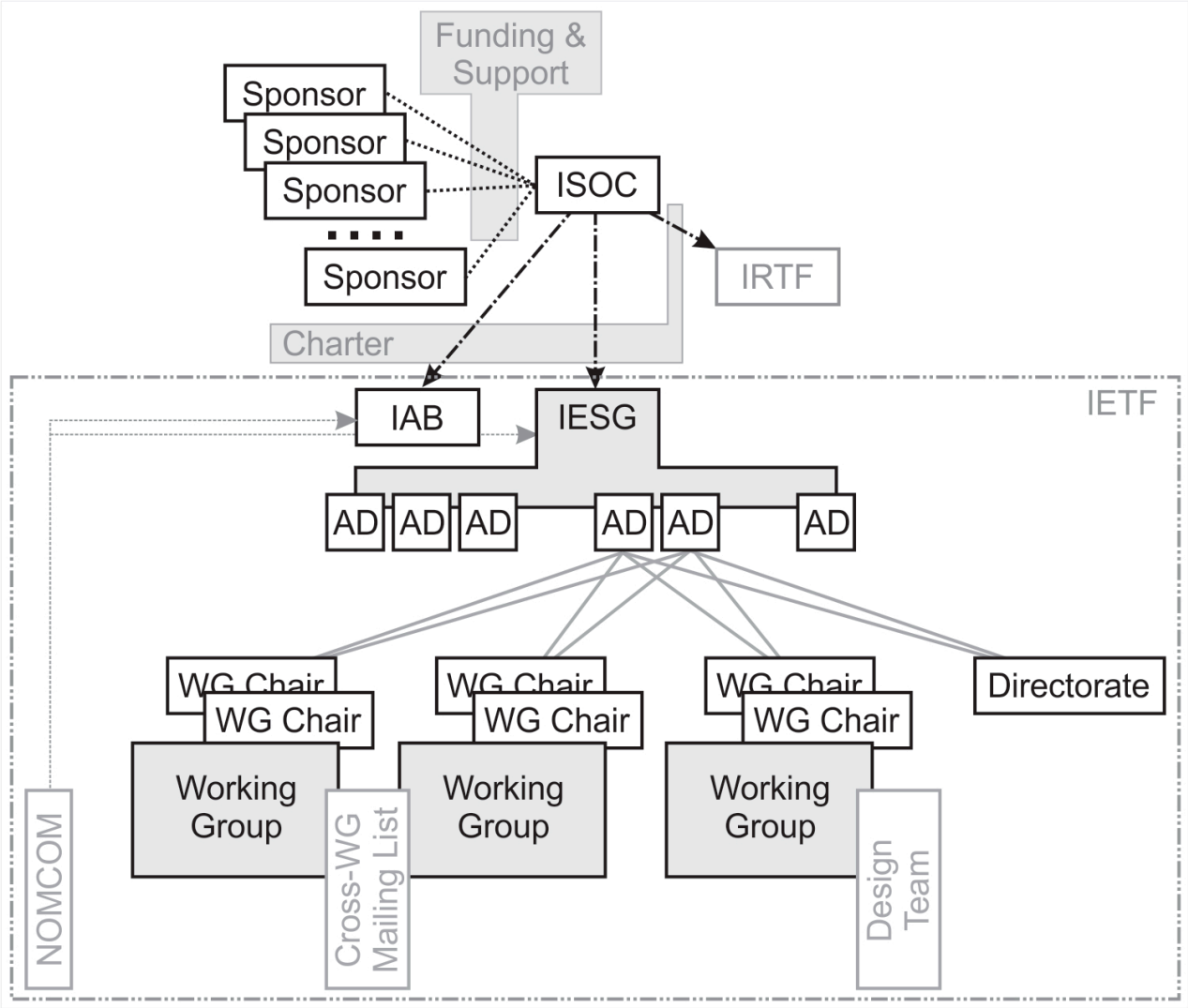


Diagram showing the organizational structure of the IETF and related entities involved in internet protocol standards development.

Reference Videos

- "What is Protocol? full Explanation | TCP/IP, HTTP, SMTP" – YouTube: https://www.youtube.com/watch?v=6T6eXk2mAx8
- "Network Protocols – ARP, FTP, SMTP, HTTP, SSL, TLS" – YouTube: https://www.youtube.com/watch?v=E5bSumTAHZE

5. TCP vs UDP

Content
TCP (Transmission Control Protocol)

- Connection-oriented, ensures reliable, ordered delivery.
- Implements flow control (windowing), congestion control.
- Header size: 20 bytes.

UDP (User Datagram Protocol)

- Connectionless, lower overhead, no delivery guarantees.
- Suitable for real-time applications.
- Header size: 8 bytes.

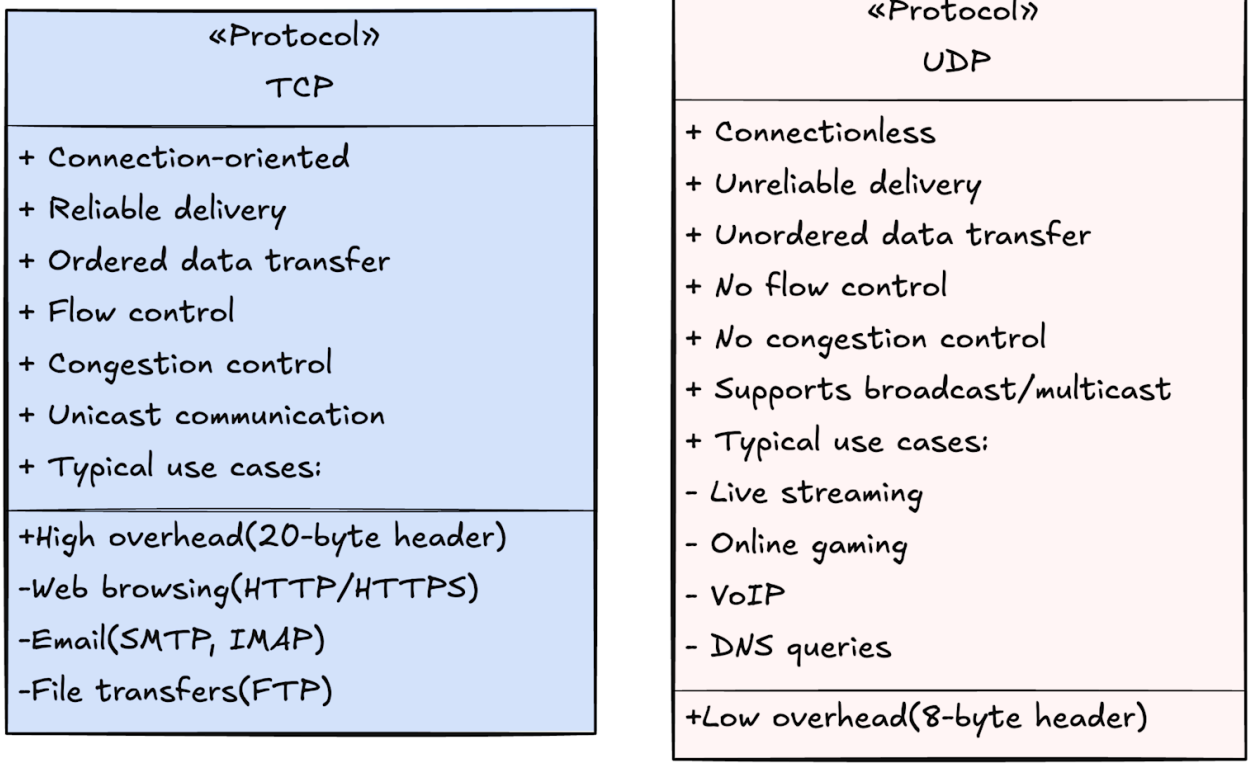
Use Cases

- **TCP:** Web browsing, email, file transfer.
- **UDP:** Live video streaming (QUIC/HTTP3), VoIP (SIP/RTP), DNS queries, online gaming.

Realtime Examples

- **Online Gaming:** "Fortnite" uses UDP to minimize latency spikes.[2]
- **VoIP:** WhatsApp calls run over UDP for smoother voice transmission.

Reference Images
Side-by-side TCP vs UDP comparison chart:



Metered.ca TURN Servers

Comparison of TCP and UDP protocols highlighting key features, typical applications, and header overhead differences.

Reference Videos

- "Network Protocols Explained: Networking Basics" – YouTube: https://www.youtube.com/watch?v=1zVZ9cWFnCc

- “Every NETWORK PROTOCOL Explained in 3 minutes” – YouTube: https://www.youtube.com/watch?v=q0S_NRq6BVc

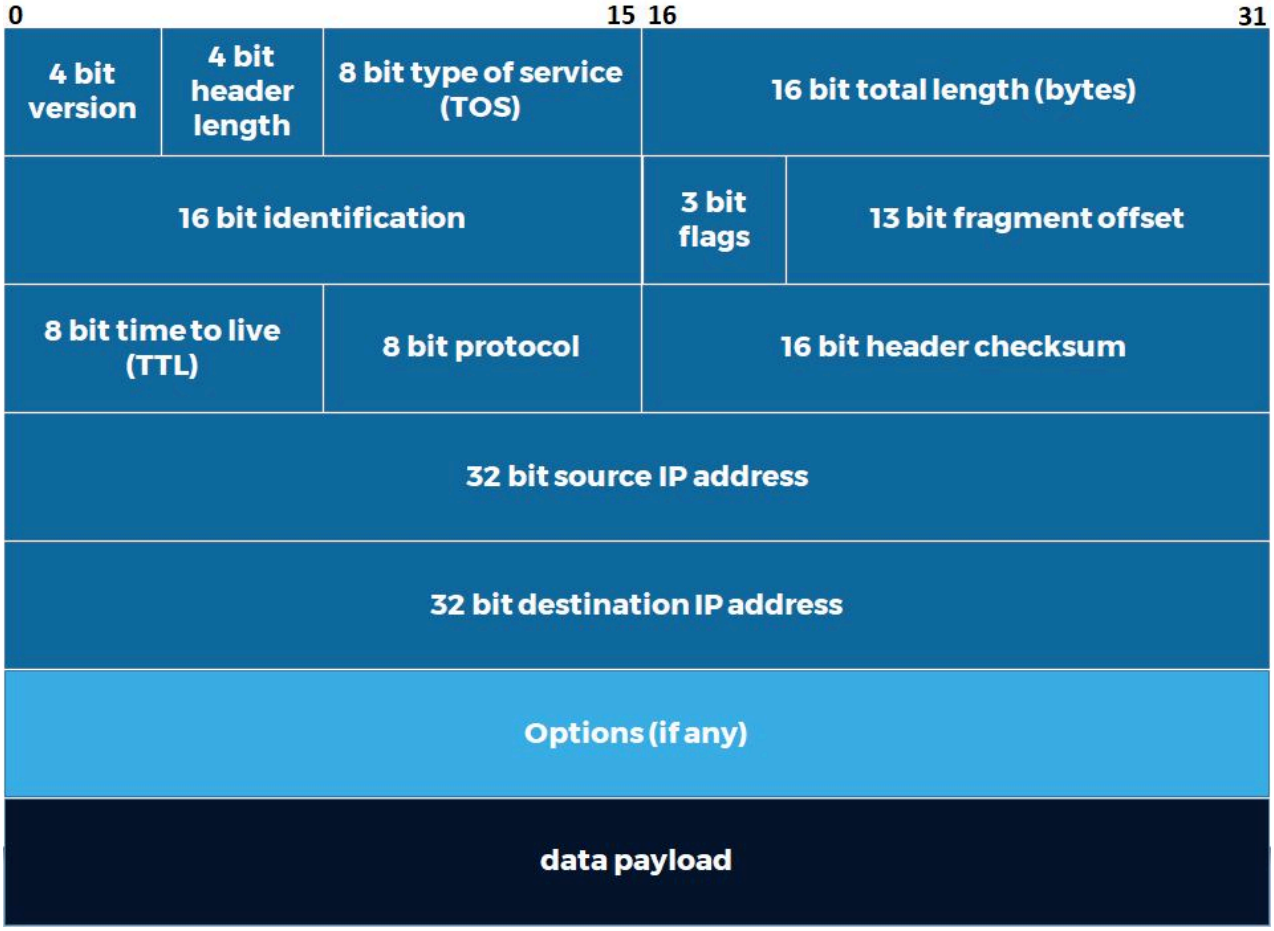
6. IP Header Structures

Content
The **IPv4 header** carries control information for packet routing:

- **Version (4 bits), IHL (4 bits), Type of Service (8 bits), Total Length (16 bits).**
- **Identification (16 bits), Flags (3 bits), Fragment Offset (13 bits).**
- **Time to Live (8 bits), Protocol (8 bits), Header Checksum (16 bits).**
- **Source IP Address (32 bits), Destination IP Address (32 bits).**
- **Options (variable).**

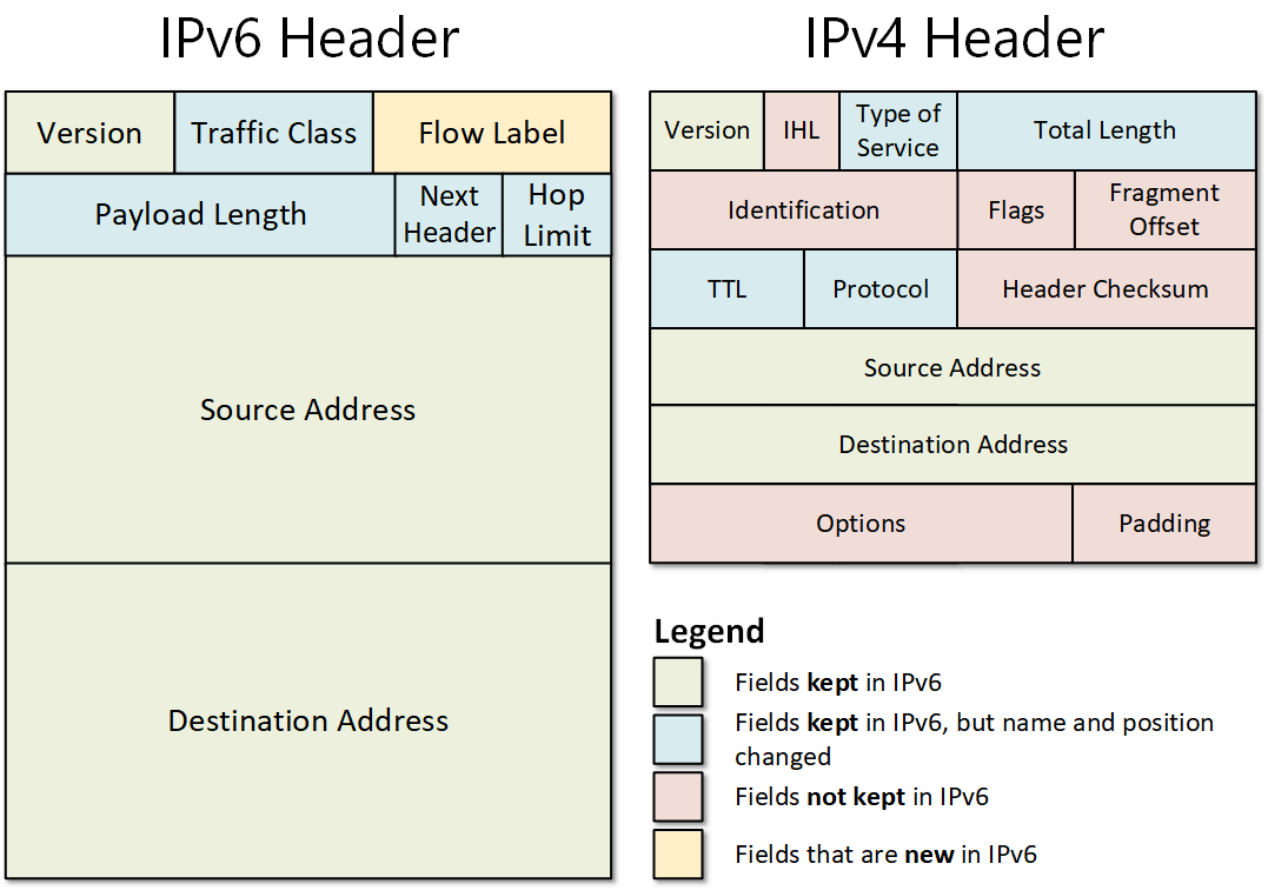
Realtime Example
IP Spoofing Attacks: In 2018, GitHub suffered a 1.35 Tbps DDoS attack leveraging forged IP header source addresses to overwhelm its infrastructure.[3]

Reference Images
IPv4 header field layout diagram:



IPv4 header structure with labeled fields and bit lengths illustrating packet components used in internet communication.

IPv4 vs IPv6 header comparison:



A comparative diagram of IPv6 and IPv4 header structures highlighting key differences and changes between the two protocols.

Reference Videos

- "How Internet Protocol (IP) Works (Networking Primer)" – YouTube: https://www.youtube.com/watch?v=eSaKz1MKsVM
- "4.3 The Internet Protocol, part 1" – YouTube: https://www.youtube.com/watch?v=sdIpLuIR1ZY

7. Exploring the Modern Internet – Deep & Dark Web

Content
The Internet's content is divided into three layers:

- **Surface Web (≈4%)**: Publicly indexed sites (Google, Wikipedia, YouTube).
- **Deep Web (≈90%)**: Unindexed content requiring authentication (banking portals, academic databases, private forums).
- **Dark Web (≈6%)**: Hidden services reachable via Tor/I2P, often associated with anonymous marketplaces and encrypted communications.

Who Uses It & How

- **Surface Web:** General public for search, commerce, social media.
- **Deep Web:** Researchers accessing subscription databases, enterprises using VPN-protected intranets.
- **Dark Web:** Whistleblowers and journalists (for anonymity), criminals trading illicit goods (e.g., Silk Road).[4]

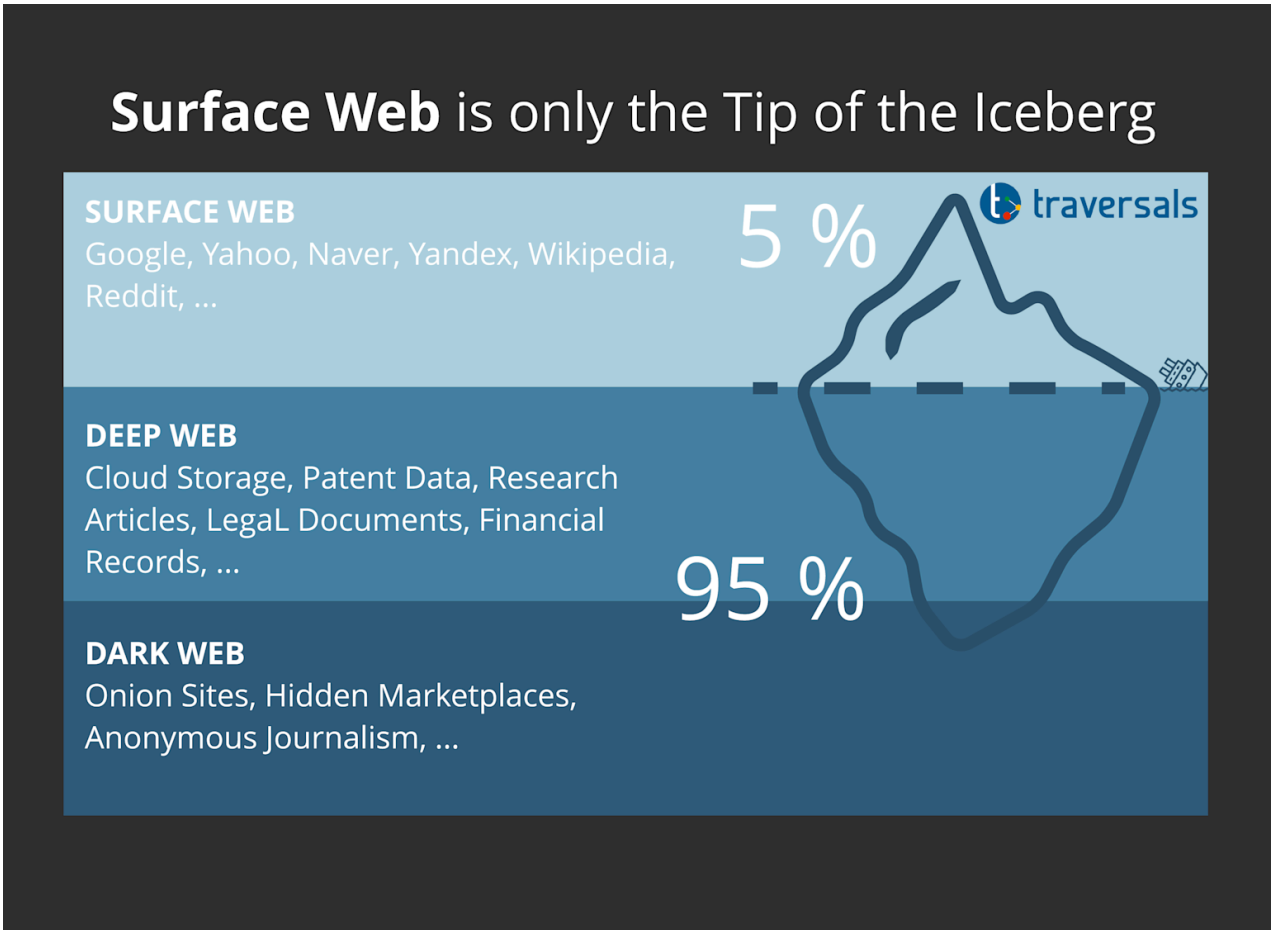
Realtime Example
Silk Road Marketplace (2011–2013) facilitated over $1.2 billion in transactions using Bitcoin before law enforcement shutdown.[4]

Reference Images
Iceberg infographic of Surface, Deep, and Dark Web:



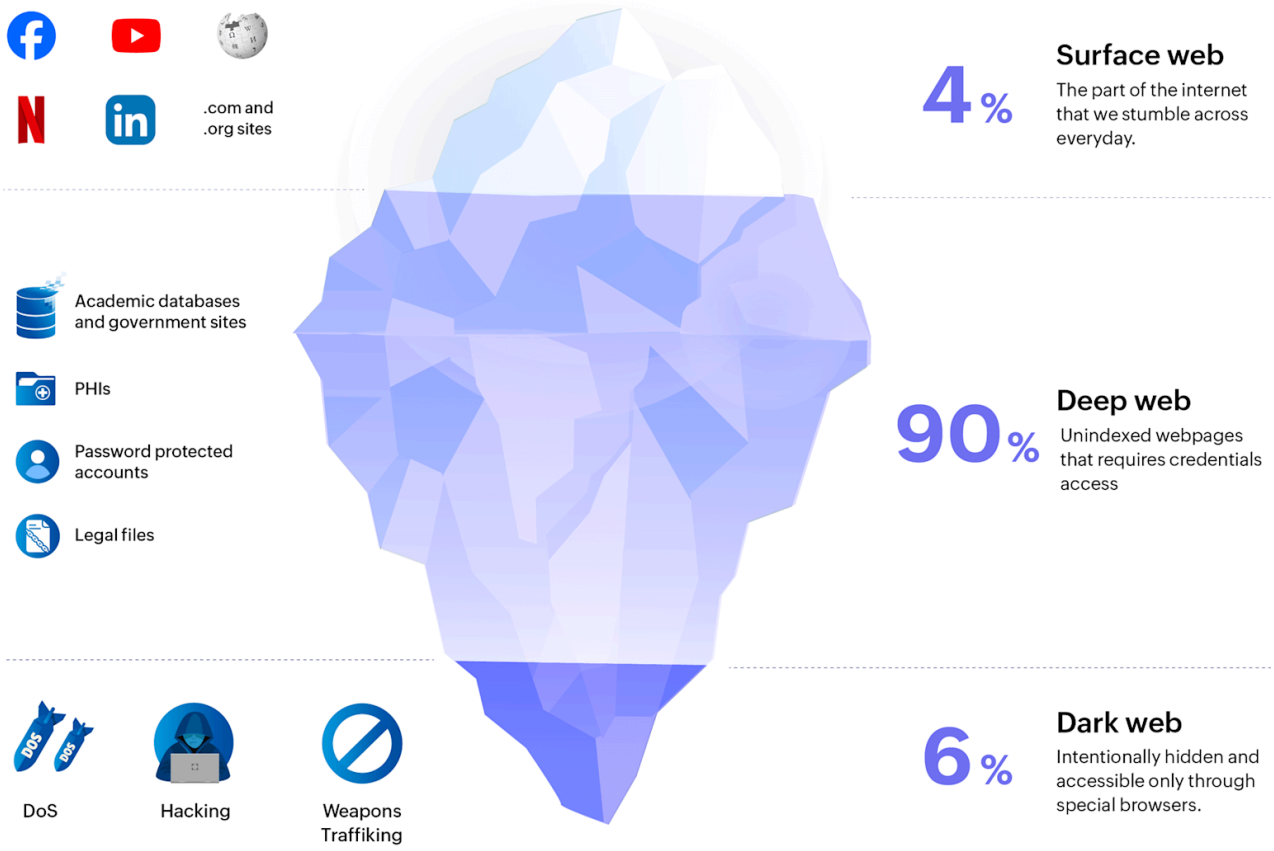
Iceberg diagram illustrating the Surface Web, Deep Web, and Dark Web with popular websites and hidden services to visualize internet layers.

Alternative iceberg diagram showing layers and examples:



Iceberg diagram showing the Surface Web as just 5% of internet content, with the Deep Web and Dark Web constituting the remaining 95%.

Detailed iceberg with legal vs illegal content:



Iceberg infographic illustrating the surface web (4%), deep web (90%), and dark web (6%) with examples of content and activities in each layer.

Reference Videos

- "Deep Web vs Dark Web: What's the Difference?" – YouTube: https://www.youtube.com/watch?v=fpqxbBEBdJM
- "The DIFFERENCE between the deep and dark web" – YouTube: https://www.youtube.com/watch?v=s4llaXGskm0

Use these images and videos to build visually rich, example-driven slides. Each section combines foundational definitions, real-world incidents, and clear diagrams to engage your audience and reinforce understanding.

⁂

1. https://www.crn.com/news/cloud/2024/facebook-instagram-outage-affects-thousands-of-users-as-meta-suffers-major-disruptions
2. https://www.wowza.com/blog/udp-vs-tcp
3. https://www.kaspersky.com/resource-center/threats/ip-spoofing
4. https://en.wikipedia.org/wiki/Silk_Road_(marketplace)